

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 08 · Straight Lines and Coordinate Geometry

Questions

Student book · questions with writing space

- 52 pages
- Past papers up to June 2026

Straight Lines and Coordinate Geometry

Your classified route

18

questions

135

marks

4

families

11

variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Straight Lines and Coordinate Geometry

Families and variants

F01 • Linear models

V01 Fit a two-point linear model and interpret or predict

F02 • Constructing related lines

V01 Line through two points in a requested form

V02 Parallel or perpendicular line through a point

V03 Perpendicular bisector from endpoint coordinates

V04 Exact line work with surd coordinates

F03 • Coordinate intersections, shapes, distance and area

V01 Related-line intersection and perpendicular distance

V02 Coordinate polygons from vectors and diagonal midpoints

V03 Coordinate lengths and areas under geometric constraints

F04 • Straight lines with curves and regions

V01 Build line/curve boundaries and describe a region

V02 Tangent parallel to a constructed line

V03 Parameterized line-quadratic intersections with a perpendicular constraint

Linear models

WHAT THIS FAMILY ASKS

Fit and interpret $y=mx+c$ from two contextual observations.

VARIANT ROUTE

V01 Fit a two-point linear model and interpret or predict

2 Complete Classified questions

10 total marks

2. A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

- (a) find the value of a and the value of b .

(4)

2. A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

- (b) State, according to the model, the height of the tree when it was planted.

(1)

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, $\pounds V$, of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

(a) Find the value of p and the value of q .

(3)

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, $\pounds V$, of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

Exactly T years after monitoring began, the share price was £2.50

- (b) Find the value of T , according to the model, giving your answer to one decimal place.

(2)

Constructing related lines

WHAT THIS FAMILY ASKS

Find line equations from points and gradients, including exact parallel, perpendicular and perpendicular-bisector constructions in the requested algebraic form.

VARIANT ROUTE

V03 Perpendicular bisector from endpoint coordinates

V04 Exact line work with surd coordinates

2 Complete Classified questions

11 total marks

1. The line l_1 passes through the point $A(-5, 20)$ and the point $B(3, -4)$.

(a) Find an equation for l_1 giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

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1. The line l_1 passes through the point $A(-5, 20)$ and the point $B(3, -4)$.

The line l_2 is perpendicular to l_1 and passes through the midpoint of AB

- (b) Find an equation for l_2 giving your answer in the form $px + qy + r = 0$, where p , q and r are integers.

(3)

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2.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Given that

- the point A has coordinates $(-2\sqrt{3}, 5)$
- the point B has coordinates $(7\sqrt{3}, 8)$
- the straight line l_1 passes through A and B

- (a) show that the gradient of l_1 is $p\sqrt{3}$, where p is a rational constant to be found.
You must show each step of your working.

(2)

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In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

- the point A has coordinates $(-2\sqrt{3}, 5)$
- the point B has coordinates $(7\sqrt{3}, 8)$
- the straight line l_1 passes through A and B

(b) Find the equation of l_2 , giving your answer in the form $y = mx + c$, where m and c are constants.

Coordinate intersections, shapes, distance and area

WHAT THIS FAMILY ASKS

Use line equations together with midpoint, vector, distance and area facts to solve coordinate-geometry configurations.

VARIANT ROUTE

V01 Related-line intersection and perpendicular distance

V02 Coordinate polygons from vectors and diagonal midpoints

V03 Coordinate lengths and areas under geometric constraints

9 Complete Classified questions

71 total marks

5. The line l_1 has equation $3y - 2x = 30$

The line l_2 passes through the point $A(24, 0)$ and is perpendicular to l_1

Lines l_1 and l_2 meet at the point P

(a) Find, using algebra and showing your working, the coordinates of P .

(5)

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5. The line l_1 has equation $3y - 2x = 30$

The line l_2 passes through the point $A(24, 0)$ and is perpendicular to l_1

Lines l_1 and l_2 meet at the point P

Given that l_1 meets the x -axis at the point B ,

(b) find the area of triangle BPA .

(3)

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5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The line l_1 has equation

$$x - 2y + 25 = 0$$

The line l_2 passes through the origin and is perpendicular to l_1

(a) Find an equation for l_2

(2)

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5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The line l_1 has equation

$$x - 2y + 25 = 0$$

The line l_2 passes through the origin and is perpendicular to l_1

The lines l_1 and l_2 intersect at the point P .

(b) Use algebra to find the coordinates of P .

(3)

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5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The line l_1 has equation

$$x - 2y + 25 = 0$$

The line l_2 passes through the origin and is perpendicular to l_1

(c) Hence find the shortest distance from l_1 to the origin.

Write your answer as a fully simplified surd.

(2)

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2. The line l_1 has equation $y = 10 - 3x$

The line l_2

- is perpendicular to l_1
- intersects l_1 at the point P with x coordinate 4

(a) Find an equation for l_2 writing the answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

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2. The line l_1 has equation $y = 10 - 3x$

The line l_2

- is perpendicular to l_1
- intersects l_1 at the point P with x coordinate 4

Given that

- line l_1 cuts the y -axis at the point M
- line l_2 cuts the y -axis at the point N

(b) find the area of triangle MPN , making your method clear.

(3)

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

- (a) Find the equation of l_2 , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

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6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

(b) the area of $PQRS$,

(3)

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The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

(c) the coordinates of S .

(2)

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8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

(a) Find the gradient of l_1

(1)

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8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

Given that

- the point A has coordinates $(6, -2)$
 - the line l_2 passes through A and is perpendicular to l_1
- (b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The line l_1 has equation

$$2x - 5y + 7 = 0$$

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

The lines l_1 and l_2 intersect at the point M .

(c) Using algebra and showing all your working, find the coordinates of M .

(Solutions relying on calculator technology are not acceptable.)

(3)

The line l_1 has equation

$$2x - 5y + 7 = 0$$

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

Given that the diagonals of a square $ABCD$ meet at M ,

(d) find the coordinates of the point C .

(2)

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2. The points P , Q and R have coordinates $(-3, 7)$, $(9, 11)$ and $(12, 2)$ respectively.

(a) Prove that angle $PQR = 90^\circ$

(3)

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2. The points P , Q and R have coordinates $(-3, 7)$, $(9, 11)$ and $(12, 2)$ respectively.

Given that the point S is such that $PQRS$ forms a rectangle,

(b) find the coordinates of S .

(2)

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Solutions relying on calculator technology are not acceptable.

(a) Find the equation of l_1 , giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)



Solutions relying on calculator technology are not acceptable.

The straight line l_2 passes through the origin O and is perpendicular to l_1

The lines l_1 and l_2 meet at the point R as shown in Figure 1.

(4)



Solutions relying on calculator technology are not acceptable.

(c) Find the exact area of triangle OPQ .

(3)

9. Given that

- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$
- the line l_1 passes through A and B

(a) find an equation for l_1 , giving your answer in the form $px + qy + r = 0$ where p , q and r are integers to be found.

(3)

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- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point C .

(4)

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- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(c) Hence find the minimum possible area of triangle ABC .

(2)

4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

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4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

The line L_2

- passes through A and is perpendicular to L_1
- cuts the x -axis at the point C

(b) Find the coordinates of C .

(3)

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4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The point $A(-1, 8)$ and the point $B(7, 2)$ lie on the line L_1

(c) Hence find the area of triangle ABC .

(3)

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Straight lines with curves and regions

WHAT THIS FAMILY ASKS

Combine line construction with quadratic equations, tangent gradients or graph inequalities.

VARIANT ROUTE

V01 Build line/curve boundaries and describe a region

V02 Tangent parallel to a constructed line

V03 Parameterized line-quadratic intersections with a perpendicular constraint

5 Complete Classified questions

43 total marks



The straight line l passes through P and Q .

- (3)



The straight line l passes through P and Q .

The quadratic curve C passes through P and has a minimum point at Q .

(b) Find an equation for C .

(3)



The straight line l passes through P and Q .

The region R , shown shaded in Figure 2, lies in the second quadrant and is bounded by C and l only.

(c) Use inequalities to define region R .

(2)

3. (i) Solve

$$\frac{3}{x} > 4$$

(3)

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The infinite region R , shown shaded in Figure 1, lies in quadrants 2 and 3 and is bounded by C and l only.

- l has a gradient of 3
- C has equation $y = 2x^2 - 50$
- C and l intersect on the negative x -axis

(3)

5. The curve C has equation $y = f(x)$

Given that

- $f(x)$ is a quadratic expression
- the maximum turning point on C has coordinates $(-2, 12)$
- C cuts the negative x -axis at -5

(a) find $f(x)$

(4)

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The line l_1 has equation $y = \frac{4}{5}x$

Given that the line l_2 is perpendicular to l_1 and passes through $(-5, 0)$

- (b) find an equation for l_2 , writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

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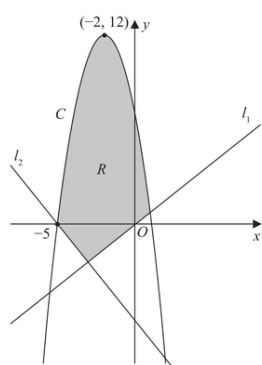


Figure 2

Figure 2 shows a sketch of the curve C and the lines l_1 and l_2

(c) Define the region R , shown shaded in Figure 2, using inequalities.

(2)

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Figure 4 shows a sketch of the curve C with equation

The point M is the maximum turning point on C .

- $$a + b(x + c)^2$$

(ii) Hence, or otherwise, state the coordinates of M .

(5)

8.

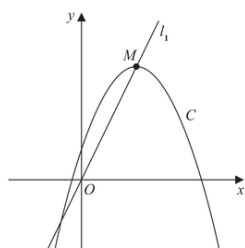


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C .

The line l_1 passes through O and M , as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

(b) Find an equation for l_2

(5)

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Figure 5 shows a sketch of the quadratic curve C with equation

The line l_1 also shown in Figure 5,

- (a) (i) Write down an equation for l_1

(1)

- (ii) Find, in terms of b , the coordinates of P

(3)

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

(b) find, in terms of b , an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

- has gradient $\frac{1}{2}$

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

$$y = -2x + \frac{5b}{2} - 4$$

(d) Hence, or otherwise, find the value of b

(2)

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